



Airfoil Near Wake Turbulent Properties at Maximum Aerodynamic Efficiency Condition

Sidaard Gunasekaran¹, Rachel Sharp²
University of Dayton, Dayton, OH 45419

From the standpoint of aerodynamic forces, maximum aerodynamic efficiency is the condition where the lift produced for a given amount of drag is at its highest. While the magnitude and the condition at which the maximum aerodynamic efficiency occurs is different for different airfoil profiles, the flow conditions that exist around the wing that brings about the maximum aerodynamic efficiency is unclear. The changes in the mean and fluctuating properties of the wake were correlated with the changes in aerodynamic efficiency to obtain insight into their relationship. Turbulent flow analysis such as statistical analysis, turbulent kinetic energy budget analysis, pressure gradient analysis, and modal analysis was performed on an experimentally obtained velocity field in the wake of a NACA 0012 airfoil. The results unanimously indicate that at the maximum aerodynamic efficiency condition, the overall wake turbulent fluctuations is at its minimum when compared to other conditions. And a significant increase in the wake turbulent production was observed soon after the maximum aerodynamic efficiency threshold is crossed.

Nomenclature

C_D	Coefficient of Drag
d	Diameter of the turbulent generator (m)
θ	Momentum thickness (m)
U_∞	Mean Freestream velocity (m/s)
u', v'	Fluctuating velocity components (m/s)
$\tilde{u}, \tilde{v}$	Instantaneous velocity components (m/s)
$\bar{U}, \bar{V}$	Mean velocity components (m/s)
U_0	Minimum wake velocity (m/s)
l_0	Wake half width (m)
$u'u', v'v', u'v'$	Reynolds stress components (m^2/s^2)
U_{RMS}, V_{RMS}	Root Mean Square of velocity components (m/s)
σ'_u, σ'_v	Standard deviation of fluctuating velocity (m/s)
ρ	Density (kg/m^3)
p'	Fluctuating pressure coefficient (Pa)
k	Turbulent kinetic energy (m^2/s^2)
ν	Kinematic viscosity (m^2/s)

I. Background

There was a widespread belief in the turbulence community that flows achieve a constant state by becoming asymptotically independent of their initial conditions. Experiments conducted by Wygnanski, Champagne, and Marasli [1] were the first to challenge this belief despite widespread acceptance that ‘turbulence forgets its origins’. In 1956, Townsend [2] used a velocity scale U_0 and a length scale (wake-half width l_0) to normalize the mean velocity and the Reynolds stress profiles in the free shear layer wake of a turbulent generator. By normalizing in this manner, it was theorized that the mean velocity and Reynolds stress of the free shear layer were independent of the geometry of the turbulent generators and streamwise location.

Gunasekaran and Altman [3], and [4] showed that the far wake free shear layer of an AR 4 flat plate is not just self-preserved and unique but also shows strong correlations with the aerodynamic efficiency of the wing. In an effort to determine if the relationship between the free shear layer and aerodynamic efficiency exists in the near wake, Steven et al. [5] performed experiments in the near wake of a SD7003 airfoil and determined that the turbulent

¹ Assistant Professor, Department of Mechanical and Aerospace Engineering, AIAA Member. gunasekarans1@udayton.edu

² Graduate student, Department of Mechanical and Aerospace Engineering, AIAA Member.

properties of the wake were at its minimum near the maximum aerodynamic efficiency condition. Upon comparison of turbulent quantities such as U_{RMS} , V_{RMS} , and Reynolds stress at different angles of attack in the near wake of SD7003, it was determined that the magnitude of these fluctuating flow quantities were at minimum near the maximum aerodynamic efficiency condition when compared to the other angles of attack.

In an effort to replicate the results as well as show that the reduction of turbulent quantities at the maximum aerodynamic efficiency condition is at minimum, experiments were conducted in the near wake of a NACA 0012 airfoil at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). It is important to acknowledge the influence of various initial conditions that could potentially affect the properties in the near wake of the airfoil such as the Reynolds number, angle of attack, airfoil geometry, trailing edge geometry, wake pressure gradients, airfoil surface roughness and freestream turbulence.

a) Influence of Reynolds number

Initial conditions such as Reynolds number play an important role in the airfoil aerodynamics, especially at low Reynolds numbers. Yarusevych, et al. [6] experimentally investigated the changes in near-wake vortex shedding and the corresponding coherent structures with respect to Reynolds number for a NACA 0025 airfoil. At low to moderate Reynolds numbers, mechanisms such as laminar separation bubble (Rodriguez et al. [7]), Kelvin-Helmholtz instability (Clinton and Browand [8]) and Tollmein and Schlichting (T-S) waves (Baines et al., [9]) plays a dominant role in determining the onset of flow separation and near wake turbulent properties.

b) Influence of Angle of Attack

Another parameter which influences the near wake properties is the angle of attack. Shan et al. [10] states that at a higher angle of attack, the formation of laminar separation bubble initiates vortex shedding in the suction side of the airfoil which in turn produces pressure disturbances that convect over the surface of the airfoil and pass over the trailing edge. The streamwise growth of the pressure disturbances results in oscillations of the separated shear layer and eventually leads to vortex shedding. With the increase in angle of attack, Sandberg et. al [11] noted that the formation of separation bubble and the onset of vortex shedding happens considerably farther upstream when compared to lower angle of attack cases.

c) Influence of airfoil geometry and trailing edge shape

It is known that an adverse pressure gradient due to curvature contributes to larger wake spreading rate and Reynolds stress. Experiments done by Hah and Lakshminarayana [12], Nakayama [13] and Weygandt and Mehta [14] confirm that even mild changes in curvature has significant effect on the mean and turbulent quantities in the wake. Zhang et. al [15] investigated the geometry effects on flow separation and near wake properties of NACA 0012 and NACA 4412 at a Reynolds number of 10,000. As expected, Zhang et. al showed conclusive evidence that increase in curvature creates greater adverse pressure gradient and leads to vortex shedding. The off-diagonal Reynolds stress contour indicated that the maximum Reynolds stress was achieved along the edges of the displacement thickness of the airfoil. However, these simulations were performed at 10-degree angle of attack for both wings, beyond the maximum aerodynamic efficiency condition of NACA 4412. Airfoils with different trailing edge geometries also tend to produce near wake of different properties. Blake [16] and Guan et al. [17] showed that airfoils with different trailing edges produce different wake signatures.

d) Influence of wake pressure gradients

Liu and Thomas [18] investigated the effects of constant adverse, favorable and zero pressure gradients on the free shear layer wake of a flat plate and found the pressure gradient demonstrated a large effect on the mean flow and largest scale energy-containing motions rather than fine scale dissipative turbulence. Another study done by Liu et al. [19] on the wake of a flat plate, showed that the wake spreading rate is enhanced, the velocity defect is reduced and the turbulence intensity and the Reynolds stress are amplified when adverse pressure gradients are imposed. On the other than, when the wake develops in favorable pressure gradient, the wake spreading rate is increased, the velocity defect decay rate is increased, and the turbulence intensity and Reynolds stress are both decreased. The experiments indicated that mean velocity components of the wake showed rapid response to the imposed pressure gradients when compared to the turbulence.

e) Influence of surface roughness and freestream turbulence

Zhang et al. [20] investigated the effects of surface roughness and freestream turbulence levels on the wake characteristics of a turbine airfoil and determined that the increased surface roughness resulted in thickening of

boundary layer, augmented turbulent transport and showed increased turbulent diffusion within the wakes as they advect downstream. It was also shown that the vortex shedding frequency and strength were decreased with increase in freestream turbulence levels. However, except for broadening of the wake widths, augmented freestream turbulence intensity levels produce only small changes to wake turbulence intensity levels.

All the factors mentioned above plays a significant role in determining the airfoil performance as well as the near wake free shear layer properties. In an effort to understand the correlations between the aerodynamic efficiency and the near wake properties, it is important to consider all of these variables. However, doing so will involve a large number of parametric variations which needed to be approached on a case-by-case basis. As mentioned earlier, the near wake properties of SD7003 showed reduction in turbulent flow properties at the maximum aerodynamic efficiency condition [5]. The results from [5] were based on experiments conducted in a water tunnel. In an effort to validate the connection between the aerodynamic efficiency and the near wake flow properties, experiments were performed in a wind tunnel with different Reynolds number and with different airfoil geometry. The Reynolds number used in [5] was around 68,000. The Reynolds number used in this investigation was 135,000. A symmetric NACA 0012 airfoil is used in the present study as opposed to a cambered airfoil (SD7003) used in [5]. It is hypothesized that similar relationship between the near wake flow properties and the maximum aerodynamic efficiency will be seen in the present study.

II. Experimental Setup

A. Wind Tunnel

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open jet configuration (Fig. 1). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections. A photo of the UD-LSWT open jet configuration is shown in Fig. 1.

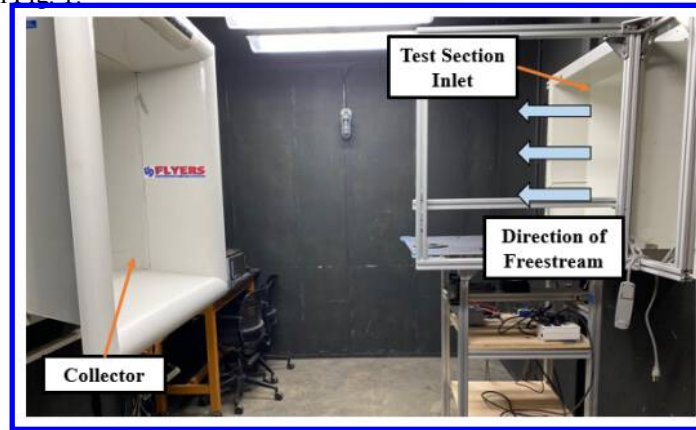


Figure 1 UD-LSWT in open-jet configuration

The test section is convertible from a closed jet configuration to an open jet configuration with the freestream range of 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at a freestream turbulence intensity below 0.1% at 15 m/s measured by hot-wire anemometer. The contraction feeds into a pressure/air sealed room where the test section is located. The effective length of the test section in the open jet configuration is 182cm (72"). A 137cm x 137cm (44" x 44") collector collects the expanded air on its return to the diffuser. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an TSI T600 Micromanometer.

C. Force-Based Experiment

The ATI Gamma Sensor was used to measure the normal force (N) and axial force (A) on the wing which were transformed into lift and drag forces which were later converted to lift and drag coefficients, C_L and C_D . The range and resolution for each direction of the sensor is shown in Table 2. The sensor was located underneath the splitter plate along with the Griffin motion rotary stage which was used to change the angle of attack. The rotary stage was controlled using Galil motion software.

Table 2 Specifications of ATI Gamma Sensor

	F_x, N	F_y, N	F_z, N	T_x, Nm	T_y, Nm	T_z, Nm
Range	65	65	200	5	5	5
Resolution	1/80	1/80	1/40	10/13333	10/13333	10/13333

D. PIV Experiment

Streamwise Particle Image Velocimetry (PIV) was conducted in the free shear layer of the NACA 0012 wing wall-to-wall model. Two splitter plates were installed at the wingtips of the wing to prevent the rollup of wingtip vortex and reduce three dimensionalities in the flow. The PIV measurements were obtained using a Vicount smoke seeder with glycerin oil and a 200 mJ/pulse Nd: YAG frequency doubled laser (Quantel Twins CFR 300). A Cooke Corporation PCO 1600 camera (1600 x 1200 pixel array) with a 105 mm Nikon lens was used to capture the images. One plano-convex lens and one plano-concave lens were used in series to convert the laser beam into a sheet. The laser and the camera were triggered simultaneously by a Quantum composer pulse generator. In each test case, over 700 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. A total of 2 iterations were performed during PIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. The streamwise PIV interrogations were conducted a Reynolds number of 135,000. The schematic of the PIV test setup is shown in Figure 2 along with the size of the field of view behind the trailing edge.

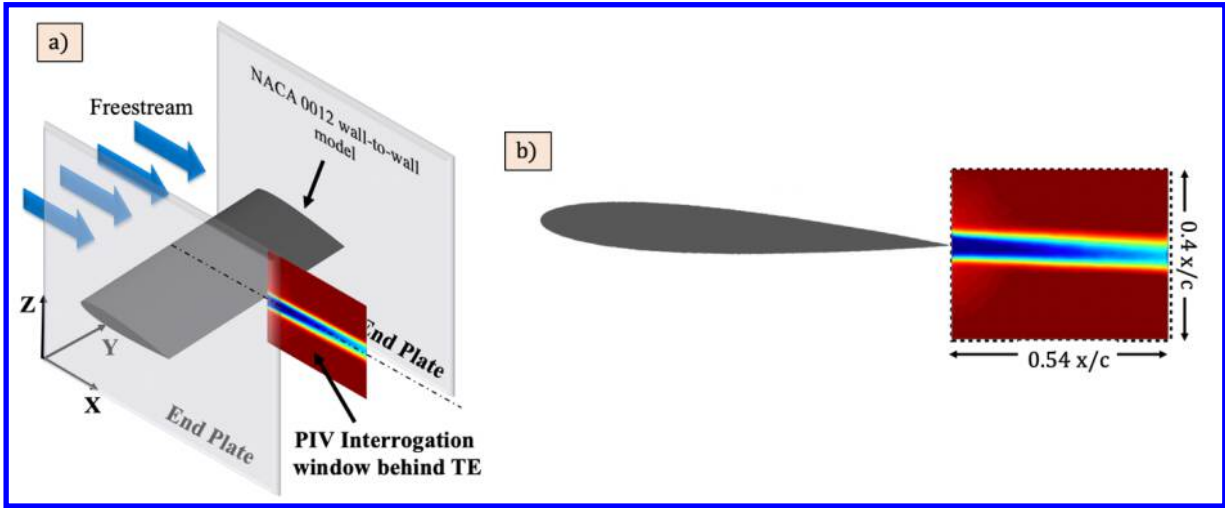


Figure 2 a) Schematic of PIV setup b) PIV Interrogation window size

III. Results

A. Statistical Approach

Due to the random and chaotic nature of turbulence flows, statistical analysis should always be relied on to determine trends in both mean and fluctuating quantities. More specifically, the probability distributions tell a compelling story behind the nature of the turbulent flows and its distributions. Different modes of statistics such as mean, standard deviation, skewness and kurtosis can be used to quantify the behavior of the turbulent flows in a broader sense. Since the exact repetition of the turbulent flow data is not possible, the probability distributions are the only way to assess both the quality of the data and its content. In this section, the cumulative distribution function and the probability density function is shown for both the instantaneous and fluctuating quantities of the wake in both streamwise and in cross-stream direction.

The CDF of the instantaneous $\tilde{u}$ and $\tilde{v}$ velocity is shown in Figure 3a and 3b respectively. The CDF shows that the data ranges from 6 m/s to 10.5 m/s. The probability of finding comparatively lower U velocities in the wake increases with increase in angle of attack as expected. The CDF of the $\tilde{v}$ velocity shows an increment in the positive v velocity with increase in angle of attack. This is an expected trend as the increased angle of attack increases the

downwash angle and thus the v velocity of the wake. The negative $\tilde{v}$ velocity in the wake is minimal as seen in the CDF. In fact, the only negative $\tilde{v}$ velocity is present at zero degree angle of attack. This could be due to the presence of vortex shedding from the trailing edge of the airfoil.

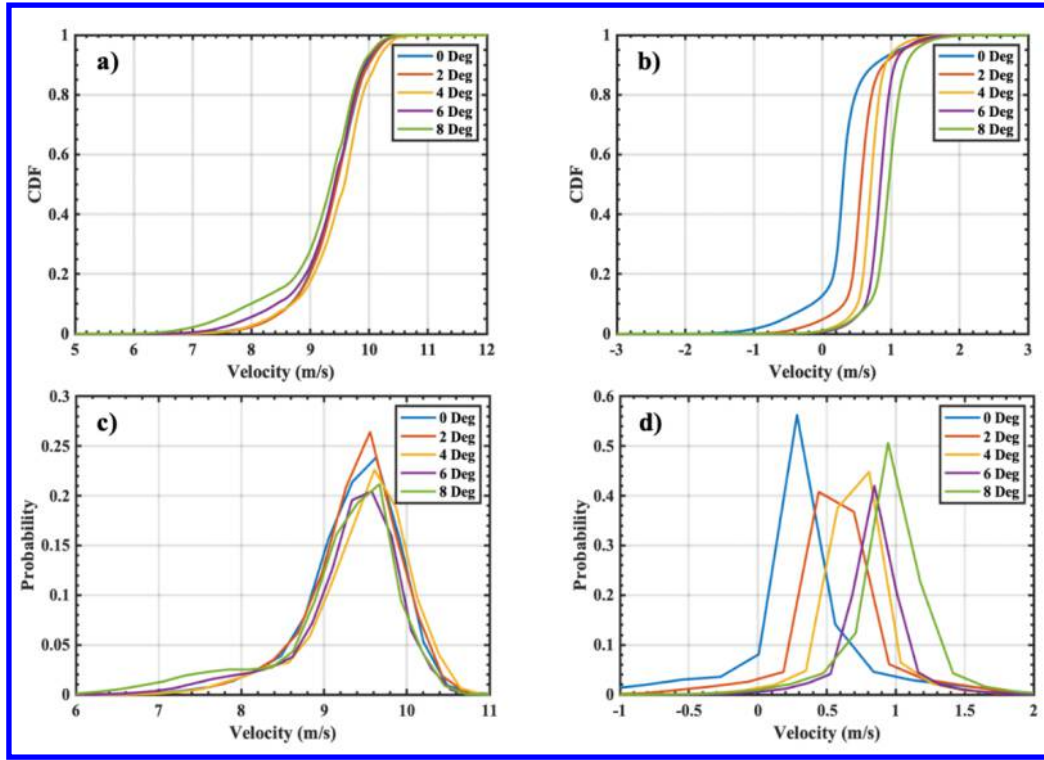


Figure 3 CDF of a) $\tilde{u}$ velocity, b) $\tilde{v}$ velocity and PDF of c) $\tilde{u}$ velocity, d) $\tilde{v}$ velocity

The corresponding PDF of the distribution which is a derivative of the CDF is shown in Figure 3c and 3d for the instantaneous $\tilde{u}$ velocity and $\tilde{v}$ velocity respectively. The number of bins was kept consistent while determining the PDF for each angle of attack cases. The PDF of the $\tilde{u}$ velocity indicates higher probability around 9.7 m/s because of the presence of the freestream velocity in larger portion of the PIV field of view. With increase in angle of attack, as seen in PDF for the 8° case, the wake grows which decreases the probability of the freestream velocity and increase the probability of lower velocities (to the left side of the distribution) which are the velocities in the wake. The PDF of the $\tilde{v}$ velocity shows higher probabilities at different v velocities in the wake with the v velocity with highest probability increasing with increase in angle of attack. Once again, this could be attributed to the increase in downwash with increase in angle of attack of the wing. It is interesting to observe that the peak probability is higher at 0° and at 8° cases when compared to the other angles of attack. This trend can be observed clearly in the CDF and PDF of the fluctuating velocities rather than the instantaneous velocities.

The CDF and PDF of the u' and v' velocities are shown in Figure 4. The CDF of the u' fluctuating velocity shows a symmetric distribution with the u' velocities ranging from -1 m/s to $+1$ m/s. This range is consistent across all angles of attack. The CDF of the v' fluctuating velocity conveys a completely different trend than that of its counterpart. The v' fluctuations range from -1.5 m/s to $+1.5$ m/s which is higher than the range of fluctuations observed in the u' fluctuating velocities. However, unlike the u' fluctuating velocity, the range of the v' fluctuating velocities decreases with increase in angle of attack. This CDF indicates that the highest v' fluctuating velocities are present at 0° angle of attack rather than at higher angle of attack. This trend is opposite to what is usually belief that the fluctuations increases with increase in angle of attack.

The corresponding PDF of the fluctuating velocities are shown in Figure 4c and 4d. The highest probability of u' fluctuating velocity occurs at 0 m/s. This is expected because the freestream generally contains much less streamwise velocity fluctuations when compared to the wake of the wing. Hence, the freestream occupies majority of the area in the field of view, the probability of u' fluctuating velocity of 0 m/s is higher. However, it is interesting to note that the fluctuations with the least probability is found at 6° angle of attack when compared to other angle of attack cases. Similar trend is also observed in the PDF of v' fluctuating velocity as shown in Figure 2d. The PDF of

the v' fluctuating velocity collapses for all the angles of attack except for the 6° angle of attack cases which has a lower probability than the other angles of attack. The range of fluctuations also appear to be slightly reduced at 6° when compared to other cases as well. It so happens that the NACA 0012 airfoils maximum aerodynamic efficiency condition occurs at 6° angle of attack.

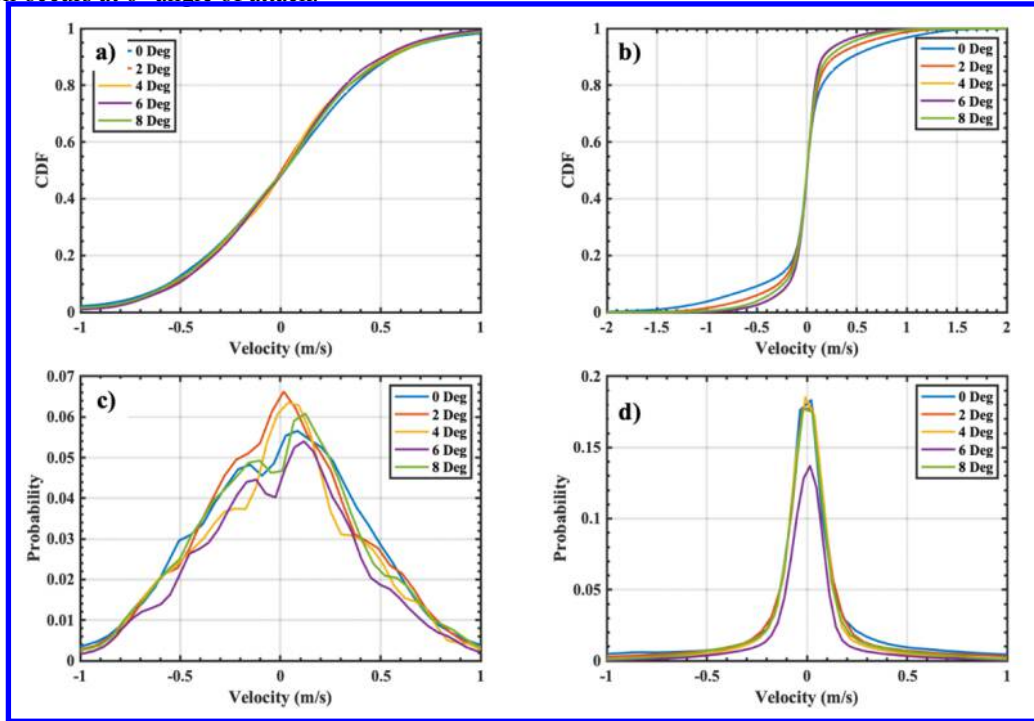


Figure 4 CDF of a) u' velocity, b) v' velocity and PDF of c) u' velocity, d) v' velocity

The CDF and PDF of the near wake data is the first indicator that the nature of the wake itself changes at the maximum aerodynamic efficiency condition with decreased velocity fluctuations when compared to other conditions. This phenomenon can further be observed in the standard deviation contour plot of the wake of the wing. The contour plot of standard deviations is shown in Figure 5.

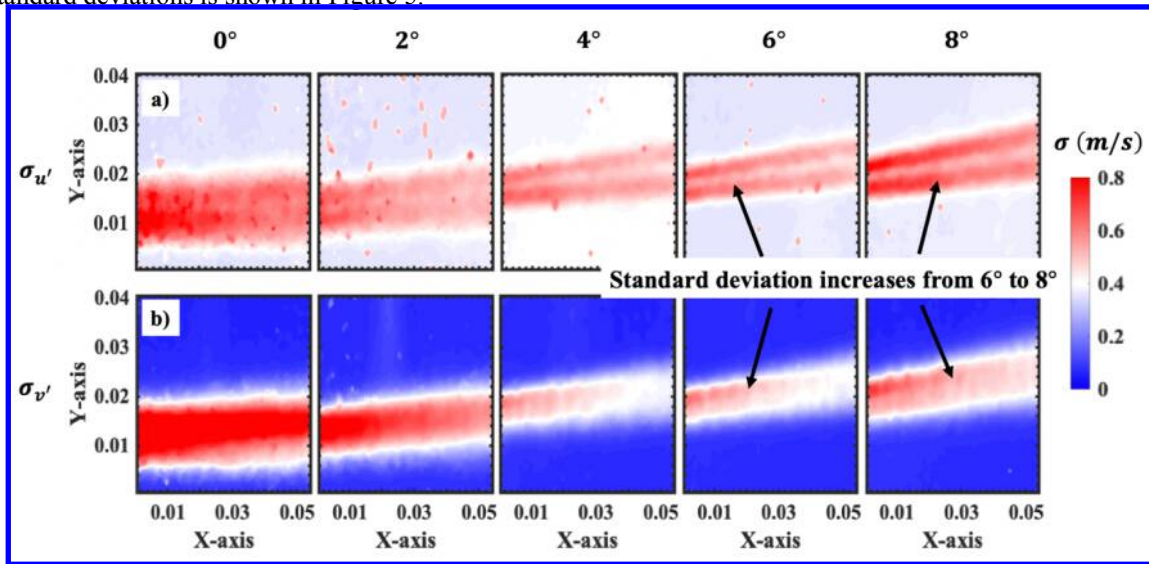


Figure 5 Standard deviation contour of u' and v' fluctuating velocity components at different angles of attack.

The contour plot was generated by taking the standard deviation of the fluctuating velocity at each and every x, y location in the field of view. It can be seen in Figure 5 that the u' standard deviation contour shows smaller σ in the

freestream and comparatively higher σ in the wake as expected. The σ at 0° angle of attack is higher and reduces with increase in angle of attack until 6° . The relatively higher standard deviation at 0° angle of attack could be due to the presence of more prominent vortex shedding in the wake. The σ increases from 6° to 8° angle of attack as indicated by the contour. This behavior is seen more prominently in the v' standard deviation shown in Figure 5b. As seen in the u' velocity standard deviation, the σ is higher at 0° and decreases with increase in angle of attack until 6° . The standard deviation of v' velocity also indicates the decrement in the wake thickness with increase in angle of attack until 6° angle of attack. The wake thickness increases going from 6° to 8° angle of attack. The wake width will be quantified using the appropriate definition later in the paper. To summarize, the statistical analysis indicates that the velocity fluctuations, along with its standard deviation are decreased at the angle of attack corresponding to the maximum aerodynamic efficiency condition.

B. Reynolds Stress Components

The Reynolds stress tensor represents the contribution of the turbulent motion to the mean stress tensor. The Reynolds stress is symmetric where the diagonal components ($u'u', v'v'$) represents normal stresses and the off-diagonal terms ($u'v', v'u'$) represents the shear stress. The normal stresses contribute very little to the transport of mean momentum whereas the off-diagonal terms play a dominant role in the mean moment transfer by turbulent motion. The normal components of the Reynolds stress tensor of the free shear layer wake is shown in Figure 6. The normal stresses are seen to be higher at 0° angle of attack and decreases until 6° , beyond which an increasing trend is seen. This decreasing fluctuation trend is similar to what is observed in the results from statistical analysis.

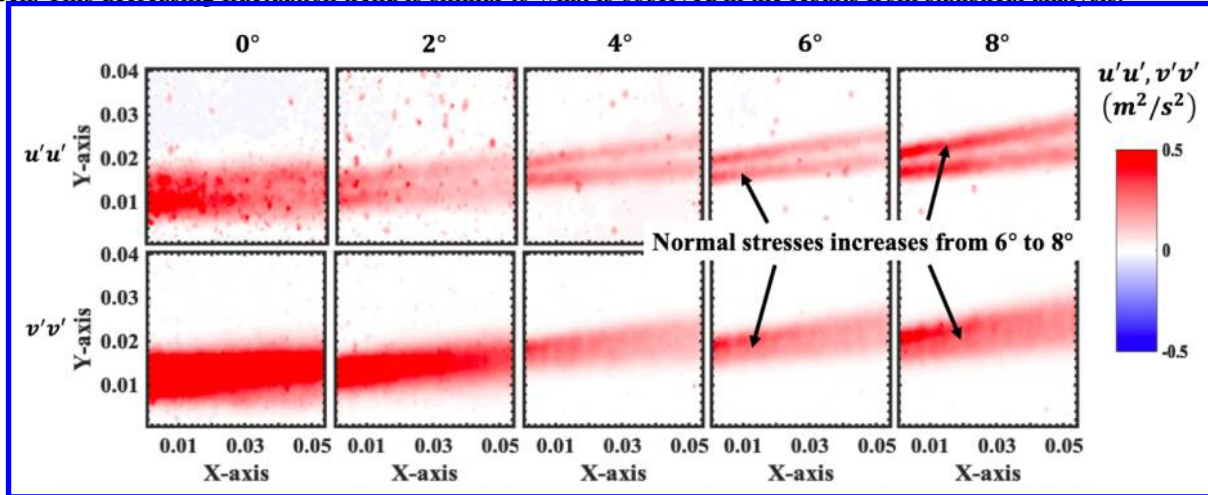


Figure 6 Contours of Reynolds normal stress components in the free shear layer wake at different angles of attack

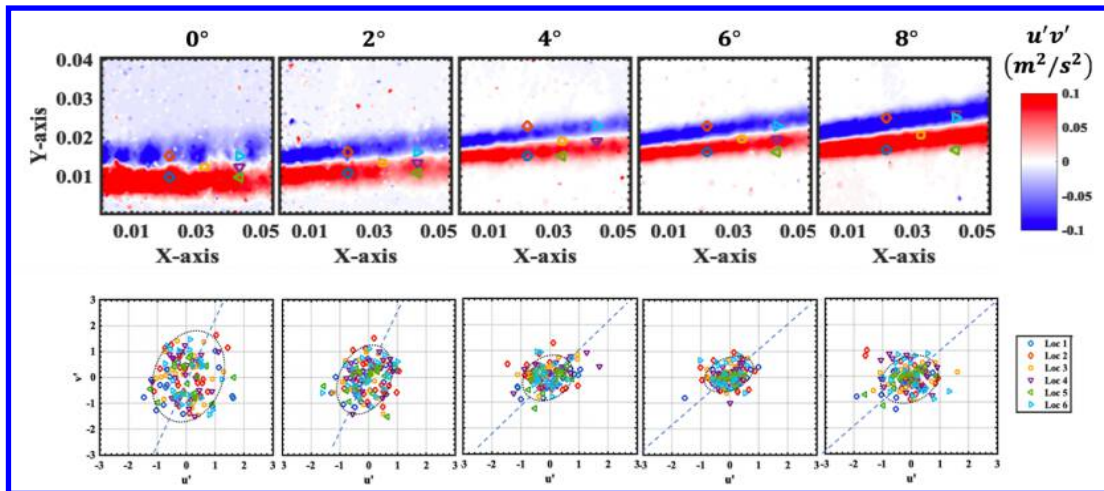


Figure 7 Contours of Reynolds shear stress in the free shear layer at different angles of attack

The Reynolds shear stress distribution in the near wake of NACA 0012 airfoil is shown in Figure 7. As previously stated, the Reynolds shear stress is major contributor to turbulent momentum transfer as it indicates the correlation between the fluctuating velocities in the two directions. The contours provide subtle differences between the magnitudes of Reynolds stress at different angles of attack which can be magnified by selecting six different locations in the free shear layer and plotting the corresponding values of u' and v' at those location. Six different points in the near wake chosen to display the correlation between the fluctuating velocities is shown in Figure 7b. It can readily be observed that the magnitude of the fluctuating velocities follows a decreasing trend until 6° angle of attack and then increases. Once again, it so happens that the maximum aerodynamic efficiency of NACA 0012 wing occurs at an angle of attack of 6° .

C. Turbulent Kinetic Energy Budget

The turbulent kinetic energy transport equation, valid for incompressible flow is given by Equation 2 [21]. The equation is only applicable to turbulent flows which are homogenous, an assumption made in the current analysis.

$$\underbrace{\frac{Dk}{Dt}}_{\text{Material Derivative of TKE (LHS)}} = \underbrace{-\frac{\partial}{\partial x_i} \overline{u'_i \left(\frac{p'}{\rho}\right)}}_{\text{Pressure Diffusion (RHS Term 1)}} - \underbrace{\frac{\partial}{\partial x_i} \overline{u'_i k'}}_{\text{Turbulent Diffusion (RHS Term 2)}} - \underbrace{\overline{u'_i u'_j} \frac{\partial \bar{U}_j}{\partial x_i}}_{\text{Turbulent Production (RHS Term 3)}} + \underbrace{\nu \frac{\partial^2 k}{\partial x_i \partial x_i}}_{\text{Viscous Diffusion (RHS Term 4)}} - \underbrace{\nu \frac{\partial u'_j}{\partial x_i} \frac{\partial u'_j}{\partial x_i}}_{\text{Dissipation (RHS Term 5)}} \quad (2)$$

where $k = 1/2 (\overline{u'_i u'_i})$ is the mean turbulent kinetic energy per unit mass, and $k' = 1/2 (\overline{u'_i u'_i})$ is the fluctuating turbulent kinetic energy per unit mass. For a 2D flow, Equation 2 simplifies to the following components:

Convection:

$$-\bar{U} \frac{\partial k}{\partial x} - \bar{V} \frac{\partial k}{\partial y} \quad (3)$$

Pressure Diffusion:

$$-\frac{\partial}{\partial x} \overline{u' \frac{p'}{\rho}} - \frac{\partial}{\partial y} \overline{v' \frac{p'}{\rho}} \quad (4)$$

Turbulent Diffusion:

$$-\frac{\partial}{\partial x} \frac{1}{2} \overline{(u'^3 + u' v'^2)} - \frac{\partial}{\partial y} \frac{1}{2} \overline{(u'^2 v' + v'^3)} \quad (5)$$

Turbulence Production:

$$-(\overline{u'^2} - \overline{v'^2}) \frac{\partial \bar{U}}{\partial x} - \overline{u' v'} \left(\frac{\partial \bar{U}}{\partial y} + \frac{\partial \bar{V}}{\partial x} \right) \quad (6)$$

Viscous Diffusion

$$\nu \frac{\partial^2 k}{\partial x^2} + \nu \frac{\partial^2 k}{\partial y^2} \quad (6)$$

Dissipation:

$$-\nu \left[\overline{\left(\frac{\partial u'}{\partial x} \right)^2} + \overline{\left(\frac{\partial u'}{\partial y} \right)^2} + \overline{\left(\frac{\partial v'}{\partial x} \right)^2} + \overline{\left(\frac{\partial v'}{\partial y} \right)^2} \right] \quad (7)$$

The first term on the LHS of Equation 2 represents the rate of change of turbulent kinetic energy. The second term in LHS represents the transport of turbulent kinetic energy by convection. The first term on the RHS represents the production of turbulent kinetic energy by interaction of Reynolds stress and the mean flow. The second on the RHS of Equation 2 represents the transport of kinetic energy by pressure. The third term on the RHS represents the

transport of turbulent kinetic energy by Reynolds stress. The fourth term represents the transport of kinetic energy by viscous stresses and the last term represents the rate of dissipation. All the terms in Equation 1 except for the pressure diffusion was obtained from PIV data. The pressure diffusion in the TKE could not be directly measured. Hence, it was estimated by equating the left hand side and the right hand side of the TKE equation. The resultant variation of each term in the TKE budget is shown in Figure 8.

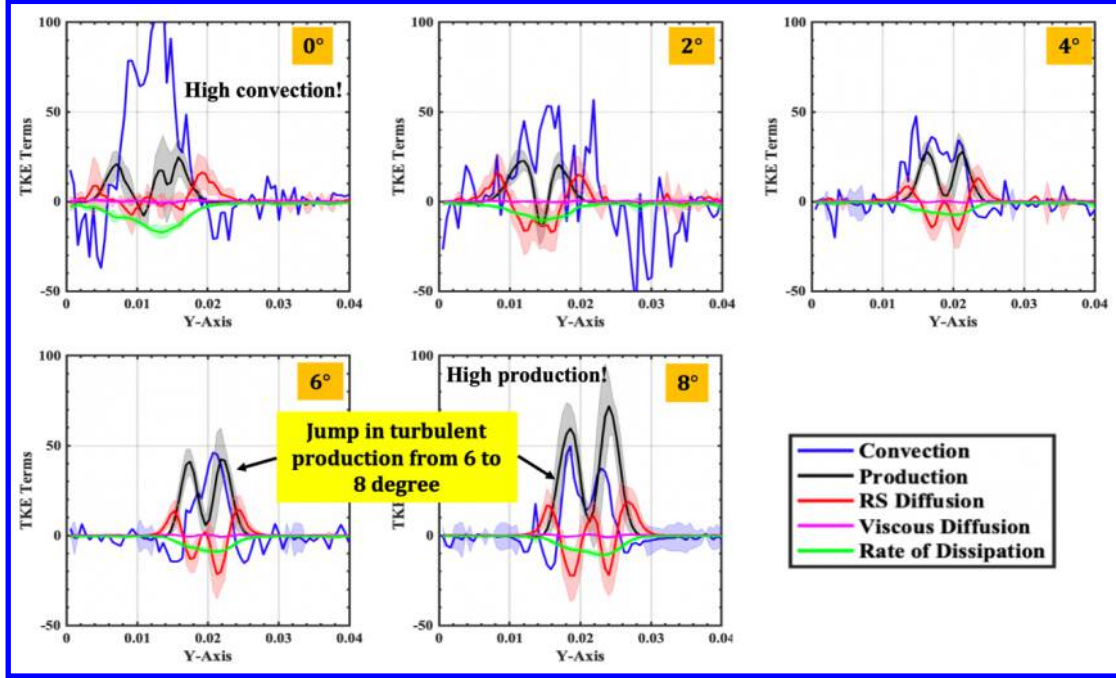


Figure 8 TKE Budget analysis at different angles of attack in the near wake of NACA 0012

At 0° angle of attack, the turbulent convection has a greater magnitude than the rest of the terms. Once again, this could be due to vortex shedding in the wake. The shaded region in the profiles represents the standard deviation of each term. The uncertainty in the turbulent convection was relatively high and therefore has not been indicated in the figure. While the turbulent convection is high at 0°, the rest of the terms, the turbulent production, Reynolds stress diffusion and rate of dissipation is comparatively minimal. As expected, the viscous diffusion doesn't play a significant role in the transport of turbulent kinetic energy. With increase in angle of attack, the Reynolds stress diffusion trend begins to take form with a positive diffusion at the edges of the wake and the negative diffusion in the center of the wake. This indicates at the edges of the wake, diffusion occurs in the downstream direction and in the center of the wake, diffusion occurs in the upstream direction. The jet-like profile of the turbulent dissipation indicates most of the dissipation occurs in the wake center rather than at the edges as expected. The production term on the other hand takes a bimodal distribution. The turbulent production occurs when the fluctuating velocities derives energy from two major components: mean shear flow and pressure gradients (dilatational production). Majority of the contribution to production comes from the former. With increase in angle of attack, the respective profiles becomes well defined. The turbulent production increases significantly from 6° to 8° angle of attack. Coincidentally, the increase in turbulent production after 6° angle of attack is also accompanied by the increase in the turbulent diffusion (diffusion due to Reynolds stresses) as well. Given that the maximum aerodynamic efficiency occurs at 6° angle of attack, the TKE analysis indicates a sharp increase in turbulent production soon after the maximum aerodynamic efficiency condition. This increase in turbulent production could be due to the onset of flow separation.

The pressure diffusion term which was calculated by determining the difference between the LHS and RHS of the TKE equation is shown in Figure 9. Rearranging Equation 2 to solve for the pressure diffusion,

$$\underbrace{-\frac{\partial}{\partial x_i} \overline{u'_i \left(\frac{p'}{\rho} \right)}}_{\text{Pressure Diffusion (RHS Term 1)}} = \underbrace{U_j \frac{\partial k}{\partial x_j}}_{\text{Convection (LHS)}} - \underbrace{\frac{\partial}{\partial x_i} \overline{u'_i k'}}_{\text{Turbulent Diffusion (RHS Term 2)}} - \underbrace{\overline{u'_i u'_j} \frac{\partial \bar{U}_j}{\partial x_i}}_{\text{Turbulent Production (RHS Term 3)}} + \underbrace{\nu \frac{\partial^2 k}{\partial x_i \partial x_i}}_{\text{Viscous Diffusion (RHS Term 4)}} - \underbrace{\nu \frac{\partial u'_j}{\partial x_i} \frac{\partial u'_j}{\partial x_i}}_{\text{Dissipation (RHS Term 5)}} \quad (8)$$

The pressure diffusion term is denoted as the “difference” in Figure 9. At 0° angle of attack, the high convection of turbulent transport is balanced by the high magnitude of downstream (positive) pressure diffusion. The magnitude of the rest of the terms in RHS is comparatively lower. With increase in angle of attack, the trends change. The RHS of the TKE equation begins to take a bimodal distribution, primarily driven by turbulent production. Whereas the TKE convection reduces along with the pressure diffusion. At 6° angle of attack, the pressure diffusion occurs in the upstream direction at the edges of the wake whereas a positive downstream pressure diffusion occurs in the center of the wake. However, after 6° angle of attack, majority of the pressure diffusion occurs in the upstream direction rather than in the downstream direction as indicated by the negative pressure diffusion term.

In order to understand the relationship between the turbulent pressure diffusion and aerodynamic efficiency, the pressure diffusion profile shown in Figure 9 is integrated along the width of the wake and the resultant variation is shown in Figure 10 as a function of angle of attack.

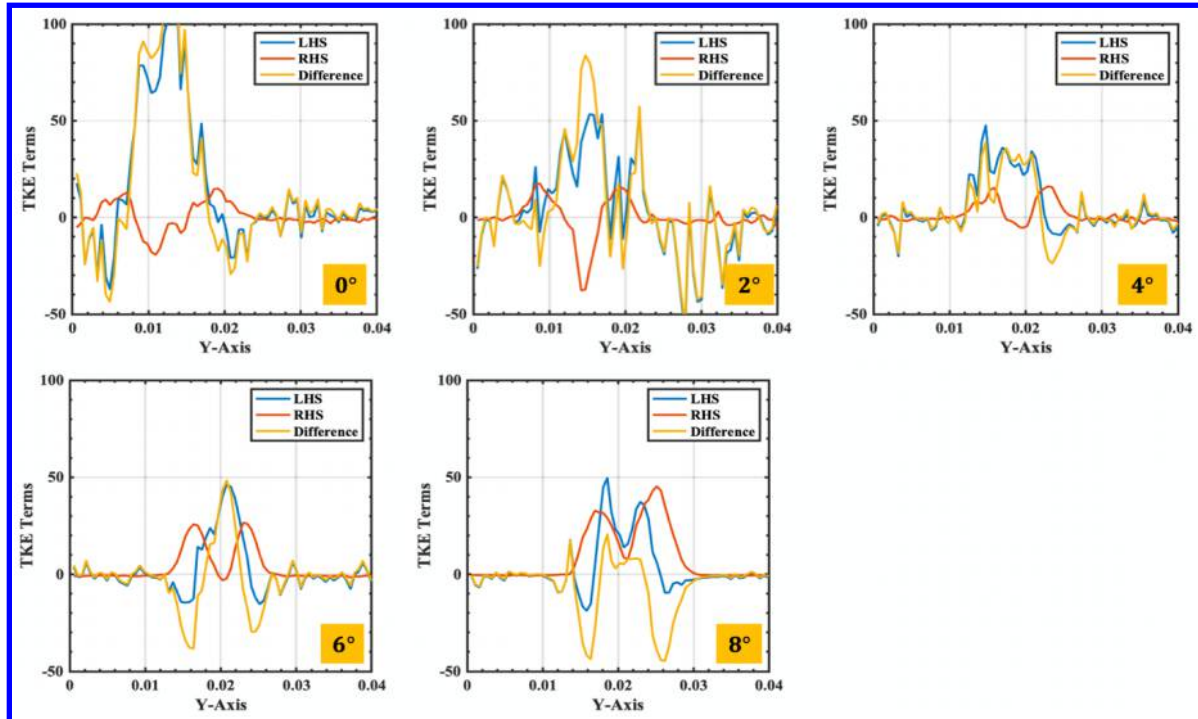


Figure 9 Balance of LHS and RHS of TKE equation

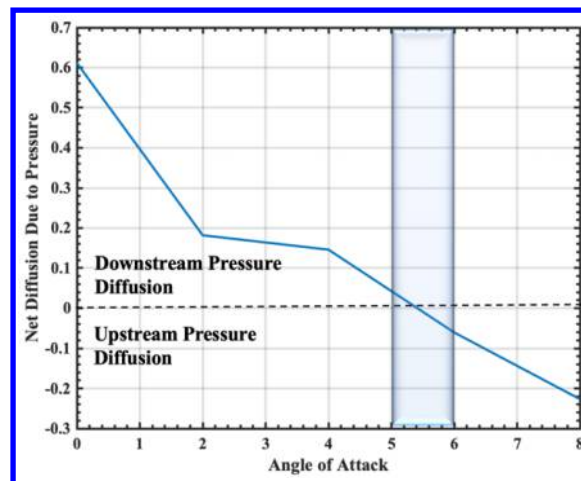


Figure 10 Net diffusion due to pressure in the wake of the wing determined by integrating the pressure diffusion profile shown in Figure 9.

It can be seen in Figure 10 that until about 6° angle of attack, the net diffusion due to pressure is in the downstream direction. However, after crossing the 6° angle of attack, the net diffusion due to pressure becomes negative which indicates an upstream diffusion due to pressure. Figure 10 indicates that the maximum aerodynamic efficiency condition is where adverse pressure diffusion begins to occur.

D. Modal Analysis

To further elucidate the changes in the near wake at the maximum aerodynamic efficiency condition, modal analysis was done in the V component of the velocity. The variation of percent energy with the number of modes is shown in Figure 11. The highest energy content is seen in the first two modes in the 0° angle of attack case succeeded by the 2° angle of attack case. The variation of 4° , 6° and 8° angle of attack cases show similar variation in the modal energy. The presence of high modal energy at 0° and 2° angles of attack is a strong indicator of vortex shedding at these angles of attack.

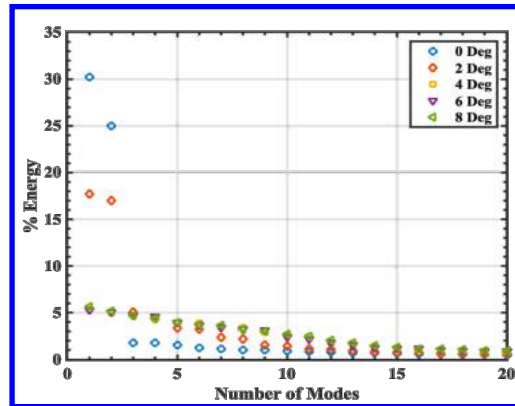


Figure 11 Variation of percent energy with number of modes for V velocity component in the wake at different angles of attack.

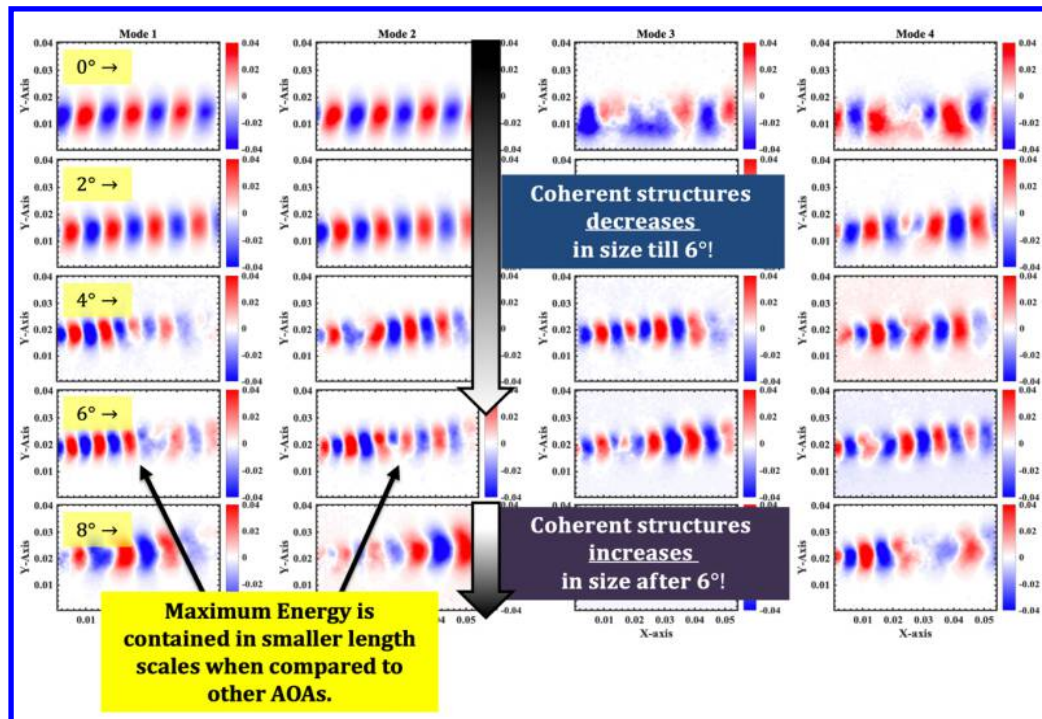


Figure 12 V velocity POD modes in the free shear layer wake at different angles of attack

The contours of different modes are shown in Figure 12 at different angles of attack. The alternating patterns in the different modes represents the coherent structures present in the wake. The period of the alternating structures is an indication of the length scales present in the wake. For all angles of attack, the higher modes consists of coherent structures with a well-defined length scale. With increase in angle of attack, the length scales of the coherent structures containing the highest energy decreases until 6° angle of attack. After 6° angle of attack however, the length scales increases as seen in Mode 1 column in Figure 9. It is interesting to note that at maximum aerodynamic efficiency condition, it is not just the velocity fluctuations which decreases but also the corresponding length scales.

IV. Conclusions

The turbulent wake signature was experimentally investigated at different angles of attack in the wake of NACA 0012 airfoil. Statistical analysis indicated that the overall fluctuations, especially in v' velocity are reduced at the maximum aerodynamic efficiency condition. Both the Reynolds normal and shear stress contours indicated the reduction in turbulent stresses at the maximum aerodynamic efficiency condition. The TKE analysis indicated a significant increase in turbulent production at angles of attack higher than the maximum aerodynamic efficiency condition. The balance of TKE equation also indicated that the turbulent diffusion due to pressure changes sign from downstream diffusion to upstream diffusion around the maximum aerodynamic efficiency condition. Finally, the modal analysis also indicated the reduction in length scales containing the highest energy at the maximum aerodynamic efficiency condition when compared to other angles of attack.

V. References

- [1] Wygnanski, I., F. Champagne, and B. Marasli. "On the large-scale structures in two-dimensional, small-deficit, turbulent wakes." *Journal of Fluid Mechanics* 168 (1986): 31-71.
- [2] Townsend, A. A. "The mechanism of entrainment in free turbulent flows." *Journal of fluid mechanics* 26, no. 4 (1966): 689-715.
- [3] Gunasekaran, S., and A. Altman. "Is there a relationship between the turbulence character of free shear layer and the wingtip vortex?." In *54th Aerospace Sciences Meeting and Exhibit, San Diego, CA*. 2016.
- [4] Gunasekaran, Sidaard. "Relationship between the free shear layer, the wingtip vortex and aerodynamic efficiency." PhD diss., University of Dayton, 2016.
- [5] Goodman, Steven, Sidaard Gunasekaran, Aaron Altman, and Albert Medina. "On the Near Wake Turbulent Flow Properties of the SD7003 Airfoil." In *AIAA Scitech 2019 Forum*, p. 0073. 2019.
- [6] Yarussevykh, S., Sullivan, P.E., and Kawall, J.G., "On Vortex Shedding from an Airfoil in Low-Reynolds-number Flows," *Journal of Fluid Mechanics*, Vol. 632, 2009, pp. 245–271.
- [7] Rodríguez, I., O. Lehmkuhl, R. Borrell, and A. Oliva. "Flow past a NACA0012 airfoil: from laminar separation bubbles to fully stalled regime." In *Direct and Large-Eddy Simulation IX*, pp. 225-231. Springer, Cham, 2015.
- [8] Winant, Clinton D., and Fred K. Browand. "Vortex pairing: the mechanism of turbulent mixing-layer growth at moderate Reynolds number." *Journal of Fluid Mechanics* 63, no. 2 (1974): 237-255.
- [9] Baines, Peter G., Sharan J. Majumdar, and Humio Mitsudera. "The mechanics of the Tollmien-Schlichting wave." *Journal of Fluid Mechanics* 312 (1996): 107-124.
- [10] Shan, Hua, Li Jiang, and Chaoqun Liu. "Direct numerical simulation of flow separation around a NACA 0012 airfoil." *Computers & Fluids* 34, no. 9 (2005): 1096-1114.
- [11] Sandberg, R. D., L. E. Jones, N. D. Sandham, and P. F. Joseph. "Direct numerical simulations of tonal noise generated by laminar flow past airfoils." *Journal of Sound and Vibration* 320, no. 4-5 (2009): 838-858.
- [12] Hah, C., and B. Lakshminarayana. "Measurement and prediction of mean velocity and turbulence structure in the near wake of an airfoil." *Journal of Fluid Mechanics* 115 (1982): 251-282.
- [13] Nakayama, A. "Curvature and pressure-gradient effects on a small-defect wake." *Journal of Fluid Mechanics* 175 (1987): 215-246.
- [14] Weygandt, James H., and Rabindra D. Mehta. "Three-dimensional structure of straight and curved plane wakes." *Journal of Fluid Mechanics* 282 (1995): 279-311.
- [15] Zhang, Wei, Wan Cheng, Wei Gao, Adnan Qamar, and Ravi Samtaney. "Geometrical effects on the airfoil flow separation and transition." *Computers & Fluids* 116 (2015): 60-73.
- [16] Blake WK (1984) Trailing edge flow and aerodynamic sound, part 1.tonal pressure and velocity fluctuations, part 2. random pressure and velocity fluctuations. Technical report DTNSRDC-83/113. David W. Taylor Naval Ship Research and Development Center, Bethesda

- [17] Guan, Y., Probsting, S., Stephens, D., Gupta, A., and Morris, S.C., "On the Wake Flow of Asymmetrically Beveled Trailing Edges," *Experiments in Fluids*, Vol. 57, 2016.
- [18] Liu, Xiaofeng, and Flint O. Thomas. "Measurement of the turbulent kinetic energy budget of a planar wake flow in pressure gradients." *Experiments in Fluids* 37, no. 4 (2004): 469-482.
- [19] Liu, Xiaofeng, Flint O. Thomas, and Robert C. Nelson. "An experimental investigation of the planar turbulent wake in constant pressure gradient." *Physics of Fluids* 14, no. 8 (2002): 2817-2838.
- [20] Zhang, Qiang, Sang Woo Lee, and Phillip M. Ligrani. "Effects of surface roughness and freestream turbulence on the wake turbulence structure of a symmetric airfoil." *Physics of fluids* 16, no. 6 (2004): 2044-2053.
- [21] Hinze, J. O. "Turbulence. McGraw-Hill Publishing Co." (1975)